KENYATTA UNIVERSITY

SCHOOL OF PURE AND APPLIED SCIENCES

DEPARTMENT OF COMPUTING AND INFORMATION SCIENCE

SCO 400 PROJECT

ITISHA SYSTEM

SUBMITTED

BY

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CHAPTER TWO: LITERATURE REVIEW

2.1. INTRODUCTION

In this chapter, we delve into the literature surrounding the proposed Itisha Enhanced Ecommerce Management System. This is generally categorized as a POS system,

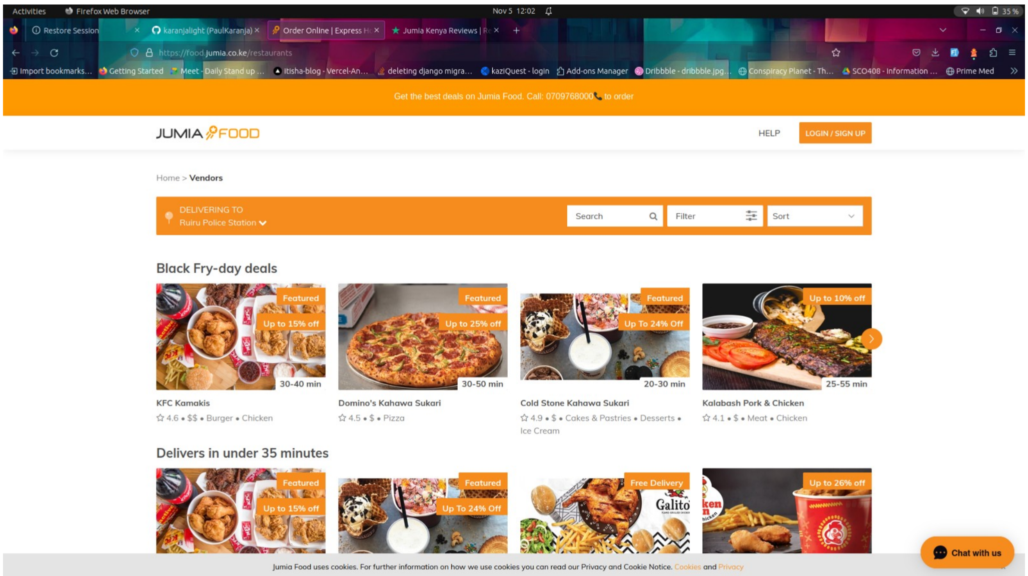
A POS system manages customer purchases, accepts payments and provides receipts. A point of sale is also where a merchant and customer conduct a retail transaction. It is where the merchant calculates the sale price for the customer, creates a record of the transaction and provides payment options.

Itisha is a POS that is web based and it's main goal is to aid in making food delivery easily accessible and storage efficient for users.

2.2 CASE STUDY 1: Organizations Similar to Itisha.

JUMIA FOOD

Jumia Food is a Kenyan Based Food delivery system that has a mobile app. It is an online food ordering site that connects people to restaurants. It also manages orders and process payments for the restaurants. It also has an extension for a rider app that facilitates order delivery to the customers.



Advantages of Jumia.

1. The mobile app and web app are user friendly.
2. Jumia offers discounts (flash sales) for a short period of time.
3. The time limit and limited availability entice consumers to buy on the spot hence driving sales.
4. Offers a reliable delivery system for the riders who deliver the meals.

Disadvantages of Jumia Food

1. In case of a failed delivery, they do not refund.
2. They do not train users, thus users do not use the system efficiently.
3. Payment systems often fail.
4. Jumia Food has poor customer service to offer after sale services.

2.3. CASE STUDY 2: Businesses related to Itisha.

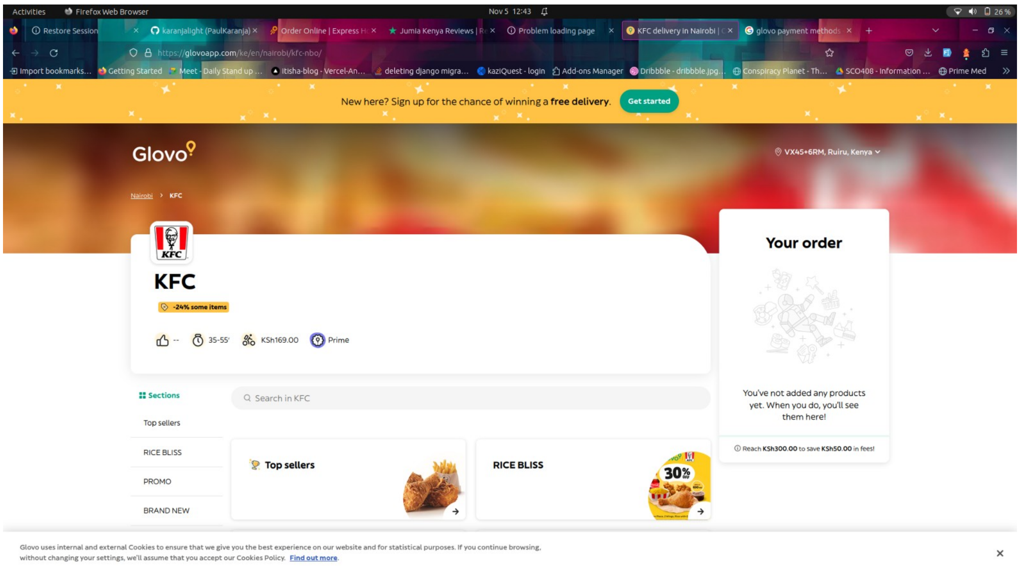
GLOVO

Glovo operates as an online grocery store, enabling customers to browse, select, and purchase a variety of products. The system handles order processing for restaurants, payment management, and real-time order tracking.

They create innovative solutions by connecting customers, businesses and couriers. It works just like Jumia Foods. Glovo and Jumia are both international companies.

Usage

The system is employed by restaurants, online grocery retailers and similar businesses looking to provide a seamless shopping experience for their customers.



Advantages

1. Glovo offers a straightforward user interface and real-time order tracking, making it a user-friendly and efficient platform. It provides comprehensive data on customer preferences and order history.
2. Glovo has efficient payment processing that can handle various methods of payments like cash, credit card and Mpesa.

Disadvantages

1. One limitation is that it may face challenges with handling peak loads during high-demand periods. Additionally, some users may require additional guidance to navigate the system effectively.

CHAPTER THREE: METHODOLOGY

3.1 APPROACH TO BE USED

The successful development of Itisha hinges on the chosen methodology. After careful consideration, we have opted for the Agile software development methodology. This methodology has been chosen for several reasons, including its suitability for dynamic and evolving projects, responsiveness to changing requirements, and its collaborative approach involving cross-functional teams.

Phases in the Agile Methodology:

1. Planning: In this phase, we outline the project objectives, define user stories, and prioritize feature development based on customer needs. This phase ensures alignment with the project's vision and goals.
2. Design: The design phase focuses on creating a user-friendly interface and defining the architecture of the "Itisha" system. Iterative design and feedback loops are key components of this phase.
3. Development: Development takes place in iterative sprints, allowing for continuous enhancements and the implementation of core features. r
4. Testing: Rigorous testing is conducted throughout the development process to ensure the system's reliability, security, and performance.
5. Deployment: Incremental deployments of the system are carried out, enabling early access for users to provide feedback and improvements.
6. Feedback and Continuous Improvement: Agile principles encourage ongoing feedback from users and stakeholders, enabling us to incorporate changes and enhancements throughout the project's life cycle.

3.2. TECHNIQUES USED TO COLLECT DATA

3.2.1. Surveys: We will design and distribute surveys to potential Itisha system users and business owners to gather valuable insights into their expectations, preferences, and challenges. Surveys will be administered both online and through targeted in-person interactions.

3.2.2. Interviews: Conducting structured interviews with business owners and potential end-users will provide an in-depth understanding of their specific needs and expectations. These interviews will help tailor the system to their unique requirements.

3.2.3. Documentation Analysis: A thorough analysis of existing documents related to ecommerce operations and management within larger businesses will be performed. This includes studying order processing records, payment methods, and existing coupon systems.

3.2.4. Observations: Observations of actual ecommerce operations and customer interactions within larger businesses will be conducted to gain practical insights into current processes and identify areas for improvement.

These data collection techniques are carefully chosen to ensure a comprehensive understanding of the project's context, end-users, and business owners. They will guide the development of the "Itisha" system to meet the specific needs of the intended audience.

3.3. TOOLS FOR IMPLEMENTING AND TESTING THE SYSTEM

The implementation and testing of Itisha Enhanced Ecommerce Management System will rely on a range of modern and robust tools, each serving a specific purpose in the development and quality assurance process. These tools include:

- Django: The backend of the system will be built using Django, a high-level Python web framework, providing rapid development capabilities and a secure foundation.

- Nuxt 3: For the frontend, Nuxt 3, a progressive JavaScript framework, will be employed to create a performant, server-rendered, and SEO-friendly user interface.

- HTML: HTML will be used for creating the structure of web pages of the ecommerce platform, ensuring compatibility and accessibility.

- Tailwind CSS: Tailwind CSS will facilitate the design and styling of the Itisha system, offering a utility-first approach to building user interfaces.

- Django Rest Framework (DRF): DRF will be utilized for developing RESTful APIs to enable seamless communication between the frontend and backend components of the system.

- Digital Ocean: The deployment of Itisha system will take place on Digital Ocean, a cloud infrastructure provider known for its scalability and reliability.

These tools will be instrumental in not only developing the system but also in conducting rigorous testing to ensure its robustness, security, and performance. Each tool plays a vital role in delivering a top-tier ecommerce management solution.

3.4. TIME SCHEDULE AND PROJECT COST

3.4.1. TIME SCHEDULE

The development of Itisha will follow a meticulously planned schedule, ensuring that each phase of the project is executed efficiently and on time. Below is a table of activities that outlines the project's timeline, with durations and respective calendar periods:

Task Name	**Duration**	**Calendar**
1. Planning	2 weeks	3th January - 17th January 2023
2. Design	2 weeks	18th June - 2th February 2023
3. Development	4 weeks	4th February – 4th March 2023
4. Testing	1 week	5th March - 20th March 2023
5. Deployment	1 week	20th March - 27th March 2023
6. Feedback and improvement	2 weeks	29th March - 10th April 2023

3.4.2. PROJECT ACTIVITY PRESENTATION

Gantt chart of activities showing graphically how we will spend time on the project.

ACTIVITY	January 2024				February 2024				March 2024				April 2024			
Weeks	1	2	3	4	1`	2	3	4	1	2	3	4	1	2	3	4
1.1 **Planning**																
2.2 **Design**																
2.3 **Development**																
2.4 **Testing**																
2.5 **Deployment**																
2.6 **Feedback**																

3.4.3. PROJECT BUDGET

NO	Item	Cost (KES)
1.	Laptop	30,000
2.	Hosting Space	10000
3.	Domain Names	2000
4.	Data Collection	5000
5.	Legal Consultation	5,000
6.	Emergency Fees	5,000
Total		57,000

3.5. REFERENCES

The research and development of the Itisha Enhanced Ecommerce Management System have drawn from a diverse range of sources, including authoritative books, peer-reviewed journals, and reputable online references. The following references provide a foundation of knowledge and insights that have contributed to the project's methodology, data collection, and tools selection.

Books:

1. Schwalbe, K. (2018). Information Technology Project Management. Cengage Learning.
2. Sommerville, I. (2016). Software Engineering. Pearson.
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1. Smith, J. D., & Anderson, P. Q. (2020). Agile software development: Best practices for large organizations. Journal of Software Engineering, 26(3), 101-118.
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3. Patel, R., & Gupta, S. (2018). Security challenges in e-commerce: A review of emerging threats and countermeasures. International Journal of Information Management, 38(1), 218-227.

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